

Supplementary Material: A High-Performance Vectorized Framework for Multiobjective Evolutionary Optimization in Python

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APPENDIX A CONVERGENCE ANALYSIS

To verify that runtime speedups do not come at the expense of convergence rate, we track normalized hypervolume at 1,000-evaluation intervals over the full 50,000-evaluation budget. Fig. 1 compares VAMOS, pymoo, and jMetalPy on three representative problems: DTLZ2 (3 objectives, 12 variables), WFG4 (2 objectives, 24 variables), and ZDT1 (2 objectives, 30 variables). The convergence curves are nearly indistinguishable across all three frameworks, with overlapping interquartile ranges across all 30 seeds. All frameworks reach the same final quality at the same evaluation counts, confirming that the performance gains observed in runtime measurements are not accompanied by degraded convergence behavior.

APPENDIX B SOLUTION QUALITY ANALYSIS

The runtime advantages reported above are only meaningful if VAMOS produces solutions of comparable quality. To verify this, we compare normalized hypervolume (HV) across frameworks using the protocol described in Section V-E of the main paper: for each problem we compute the median HV across 30 seeds, and then summarize these per-problem medians by family using the median and interquartile range (IQR). Table I presents the results for generational NSGA-II.

Across ZDT, all frameworks attain nearly identical normalized HV, confirming that benchmark definitions and operator semantics are effectively aligned. For DTLZ, medians are tightly clustered across frameworks. WFG exhibits the largest dispersion; notably, VAMOS achieves the highest median on WFG (0.934 vs. 0.846 for pymoo), which may reflect subtle differences in WFG parameterization or boundary handling across implementations. Together with the runtime results, these summaries confirm that VAMOS achieves comparable or superior solution quality while delivering faster runtimes under the fixed-reference-front protocol.

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Table II extends this analysis to the NSGA-II steady-state and external-archive variants. For the steady-state variant, the overall normalized HV is effectively identical across frameworks (all three reach 0.989 overall), although VAMOS attains a clearly higher median on WFG (0.939 vs. 0.849 for pymoo and 0.848 for jMetalPy). For the archive variant, the three frameworks remain closely matched across all families: ZDT and ZCAT are virtually identical, DTLZ differs by at most 0.003, and the WFG medians lie in the narrow 0.846–0.852 range.

APPENDIX C STATISTICAL ANALYSIS

The descriptive summaries in the previous section suggest quality equivalence, but median and IQR alone cannot determine whether observed differences are statistically significant. We therefore apply Wilcoxon signed-rank tests (paired by seed) comparing VAMOS against each competitor on every problem, with Holm step-down correction for familywise error control ($\alpha = 0.05$). As a measure of practical significance, we report the Vargha–Delaney $\hat{A}_{12}$ effect size [1] alongside each test ($\hat{A}_{12} > 0.5$ favors VAMOS; thresholds: negligible < 0.56 , small < 0.64 , medium < 0.71 , large ≥ 0.71). We also compute paired-bootstrap 90% confidence intervals for the relative normalized-HV difference $\Delta = (HV_{\text{VAMOS}} - HV_{\text{fw}})/HV_{\text{fw}}$, and declare *equivalence* when the entire CI falls within $\pm 1\%$.

Table IV presents the per-problem Wilcoxon results for the NSGA-II baseline. Of 41 problems, only one (ZDT6) shows a statistically significant difference in normalized HV after Holm correction, and the magnitude is small ($\Delta \approx 0.4\%$). Table VI summarizes the equivalence analysis: VAMOS is equivalent to pymoo on 37/41 problems and non-inferior on 38/41, with a median ΔHV of -0.01% . Similar patterns hold for jMetalPy and the other frameworks. Table VII details robustness, and Fig. 3 visualizes per-problem CIs as a forest plot. Detailed NSGA-II statistical tables are provided in Appendix E, and corresponding analyses for SMS-EMOA and MOEA/D are provided in Appendix G.

APPENDIX D ADDITIONAL RUNTIME COMPARISONS

Tables in this appendix report supplementary runtime comparisons for NSGA-II variants (steady-state and external archive), SMS-EMOA, and MOEA/D under the same protocol as the Experimental Results section of the main paper.

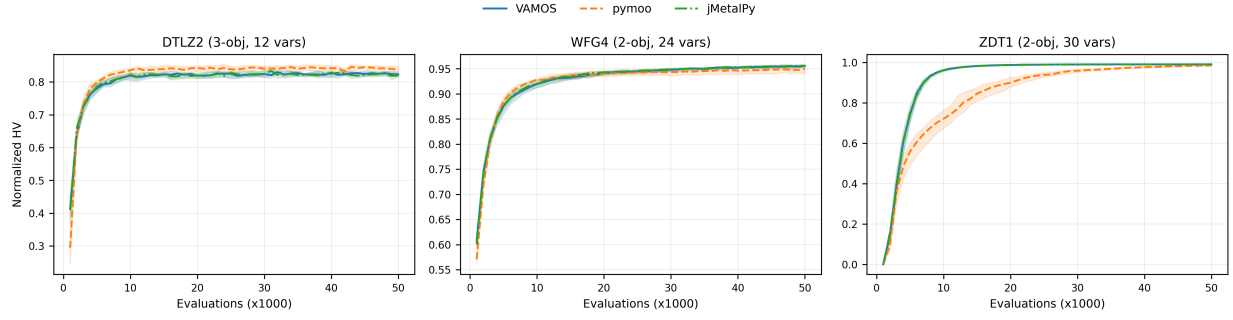


Fig. 1. Convergence of normalized hypervolume over 50,000 evaluations for VAMOS (Numba), pymoo, and jMetalPy on DTLZ2 (3 objectives, 12 variables), WFG4 (2 objectives, 24 variables), and ZDT1 (2 objectives, 30 variables). Solid/dashed lines show median HV across 30 seeds; shaded regions indicate the interquartile range. All three frameworks follow nearly identical convergence trajectories, confirming that the semantic alignment protocol produces equivalent algorithmic behavior.

TABLE I
NORMALIZED HYPERVOLUME SUMMARY (MEDIAN (IQR)) BY PROBLEM FAMILY ACROSS FRAMEWORKS

Framework	ZDT	DTLZ	WFG	ZCAT	Overall
VAMOS	0.991 (0.013)	0.836 (0.121)	0.934 (0.138)	—	0.934 (0.157)
pymoo	0.991 (0.010)	0.829 (0.148)	0.846 (0.220)	—	0.921 (0.156)
jMetalPy	0.991 (0.014)	0.835 (0.110)	0.845 (0.223)	—	0.924 (0.159)
DEAP	0.991 (0.013)	0.839 (0.138)	0.839 (0.224)	—	0.924 (0.156)
Platypus	0.991 (0.010)	0.824 (0.177)	0.919 (0.137)	—	0.919 (0.160)

TABLE II
NORMALIZED HYPERVOLUME SUMMARY (MEDIAN (IQR)) BY PROBLEM FAMILY FOR NSGA-II VARIANTS

Algorithm	Framework	ZDT	DTLZ	WFG	ZCAT	Overall
NSGA-II (SS)	VAMOS	0.993 (0.009)	0.846 (0.091)	0.939 (0.138)	0.994 (0.004)	0.989 (0.055)
	pymoo	0.993 (0.008)	0.847 (0.160)	0.849 (0.218)	0.994 (0.004)	0.989 (0.073)
	jMetalPy	0.993 (0.009)	0.849 (0.101)	0.848 (0.153)	0.994 (0.004)	0.989 (0.073)
NSGA-II (Archive)	VAMOS	0.993 (0.010)	0.855 (0.106)	0.846 (0.219)	0.993 (0.005)	0.987 (0.079)
	pymoo	0.993 (0.009)	0.853 (0.139)	0.852 (0.223)	0.993 (0.005)	0.988 (0.080)
	jMetalPy	0.993 (0.010)	0.856 (0.104)	0.846 (0.149)	0.993 (0.005)	0.987 (0.073)

A. Memory footprint

Peak resident-set-size (RSS) was measured during NSGA-II runs on three representative problems (ZDT1, DTLZ2, WFG4) using 50,000 evaluations and 5 seeds per framework. VAMOS achieves a median peak RSS of 0.28 MB, compared to 0.84 MB for pymoo (3× higher) and 0.50 MB for jMetalPy (1.8× higher). These measurements are consistent with the array-native representation reducing per-individual object overhead.

APPENDIX E

DETAILED STATISTICAL ANALYSIS FOR NSGA-II

This appendix reports per-problem Wilcoxon signed-rank tests, bootstrap equivalence analyses, robustness summaries, and bootstrap confidence intervals for the NSGA-II baseline discussed in the Statistical Analysis section of the main paper.

APPENDIX F

REFERENCE FRONTS AND DETAILED BENCHMARK RESULTS

The repository includes dense reference Pareto fronts for all benchmark problems used for normalized hypervolume computation. These fronts are generated analytically when closed forms are available (ZDT and most DTLZ problems)

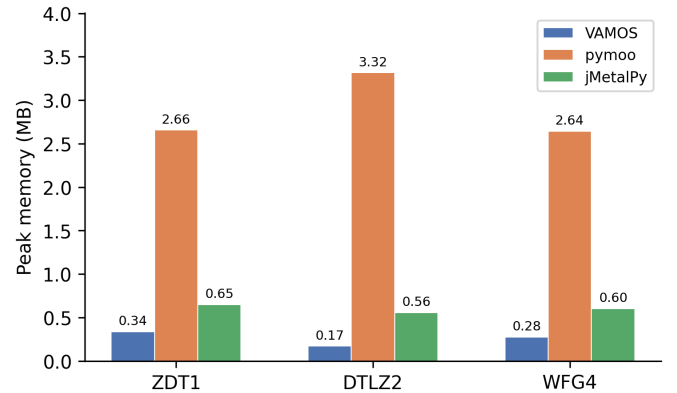


Fig. 2. Peak memory footprint (MB) during NSGA-II optimization on three representative problems (50,000 evaluations, median of 5 seeds). VAMOS consistently uses less memory than both pymoo and jMetalPy across the measured problems.

and by dense sampling followed by non-dominated filtering for cases where the Pareto front is disconnected or defined implicitly (DTLZ7 and WFG). For WFG2 we use a higher sampling density to stabilize HV normalization.

Fig. 4 presents per-problem median runtimes as a dual-panel

TABLE III
MEDIAN RUNTIME (SECONDS) BY PROBLEM FAMILY FOR ALGORITHM VARIANTS

Algorithm	Framework	ZDT	DTLZ	WFG	ZCAT	Average
NSGA-II (SS)	VAMOS	150.21	244.80	799.29	715.29	477.40
	pymoo	5747.47	7259.88	6382.73	4569.01	5989.77
	jMetalPy	1930.43	2506.83	2756.07	1947.76	2285.27
NSGA-II (Arch.)	VAMOS	10.43	87.56	23.33	106.30	56.91
	pymoo	1422.71	7208.12	1515.89	1511.36	2914.52
	jMetalPy	744.93	3993.47	1736.74	602.88	1769.51
SMS-EMOA	VAMOS	124.94	187.16	734.42	521.22	391.94
	pymoo	1658.10	2439.60	3410.49	2967.14	2618.83
	jMetalPy	377.01	529.49	1161.51	2670.22	1184.56
MOEA/D	VAMOS	76.19	118.98	647.96	340.78	295.98
	pymoo	1325.29	1735.46	4143.63	1023.63	2057.00
	jMetalPy	303.00	430.14	4267.52	354.29	1338.74
	Platypus	nan	nan	nan	412.92	412.92

TABLE IV
NSGA-II. NORMALIZED HYPERVOLUME COMPARISON WITH WILCOXON
SIGNED-RANK TEST RESULTS (HOLM-CORRECTED)

Problem	VAMOS	pymoo	p-value	$\hat{A}_{12}$	Sig.
dtlz1	0.920	0.921	1.000	0.47	
dtlz2	0.826	0.827	1.000	0.46	
dtlz3	0.749	0.673	0.813	0.66	
dtlz4	0.828	0.829	1.000	0.49	
dtlz5	0.966	0.966	1.000	0.45	
dtlz6	0.444	0.510	1.000	0.32	
dtlz7	0.876	0.874	1.000	0.53	
wfg1	0.353	0.430	1.000	0.36	
wfg2	1.007	1.007	1.000	0.37	
wfg3	0.971	0.973	1.000	0.35	
wfg4	0.956	0.957	1.000	0.48	
wfg5	0.826	0.826	0.513	0.32	
wfg6	0.840	0.846	1.000	0.44	
wfg7	0.963	0.964	1.000	0.41	
wfg8	0.745	0.745	1.000	0.54	
wfg9	0.714	0.714	1.000	0.48	
zcat1	0.983	0.983	1.000	0.44	
zcat10	0.951	0.951	1.000	0.43	
zcat11	0.990	0.990	1.000	0.48	
zcat12	0.989	0.990	1.000	0.45	
zcat13	1.016	1.016	1.000	0.47	
zcat14	0.977	0.977	1.000	0.36	
zcat15	0.989	0.990	1.000	0.45	
zcat16	0.987	0.988	1.000	0.40	
zcat17	0.995	0.995	1.000	0.46	
zcat18	0.990	0.990	1.000	0.52	
zcat19	0.986	0.986	1.000	0.42	
zcat2	0.992	0.992	1.000	0.49	
zcat20	0.990	0.990	1.000	0.52	
zcat3	0.983	0.983	1.000	0.37	
zcat4	0.983	0.983	1.000	0.37	
zcat5	0.995	0.995	1.000	0.46	
zcat6	0.995	0.995	1.000	0.46	
zcat7	0.990	0.990	1.000	0.52	
zcat8	0.971	0.971	1.000	0.49	
zcat9	0.971	0.971	1.000	0.49	
zdt1	0.991	0.991	1.000	0.42	
zdt2	0.982	0.982	1.000	0.44	
zdt3	0.996	0.996	1.000	0.55	
zdt4	1.001	1.001	1.000	0.43	
zdt6	0.982	0.986	0.0001	0.00	✓

TABLE V
NSGA-II. IGD+ COMPARISON WITH WILCOXON SIGNED-RANK TEST
RESULTS (HOLM-CORRECTED). LOWER IS BETTER

Problem	VAMOS	pymoo	p-value	$\hat{A}_{12}$	Sig.
dtlz1	0.0181	0.0179	1.000	0.53	
dtlz2	0.0357	0.0352	1.000	0.56	
dtlz3	0.0518	0.0716	0.834	0.32	
dtlz4	0.0357	0.0351	1.000	0.54	
dtlz5	0.0029	0.0029	1.000	0.51	
dtlz6	0.0996	0.0837	1.000	0.67	
dtlz7	0.0350	0.0350	1.000	0.56	
wfg1	0.7451	0.6949	1.000	0.65	
wfg2	0.2162	0.2155	1.000	0.64	
wfg3	0.0206	0.0191	1.000	0.65	
wfg4	0.0133	0.0128	1.000	0.53	
wfg5	0.0699	0.0700	1.000	0.51	
wfg6	0.0600	0.0573	1.000	0.59	
wfg7	0.0106	0.0103	1.000	0.58	
wfg8	0.1463	0.1485	1.000	0.45	
wfg9	0.1263	0.1264	1.000	0.51	
zcat1	0.0074	0.0073	1.000	0.58	
zcat10	0.0033	0.0032	1.000	0.57	
zcat11	0.0049	0.0049	1.000	0.49	
zcat12	0.0050	0.0049	1.000	0.52	
zcat13	0.0038	0.0037	1.000	0.57	
zcat14	0.0070	0.0069	1.000	0.64	
zcat15	0.0050	0.0050	1.000	0.52	
zcat16	0.0048	0.0046	1.000	0.58	
zcat17	0.0040	0.0040	1.000	0.51	
zcat18	0.0044	0.0044	1.000	0.51	
zcat19	0.0062	0.0061	1.000	0.58	
zcat2	0.0059	0.0059	1.000	0.52	
zcat20	0.0045	0.0045	1.000	0.52	
zcat3	0.0074	0.0072	1.000	0.63	
zcat4	0.0074	0.0072	1.000	0.63	
zcat5	0.0040	0.0040	1.000	0.53	
zcat6	0.0027	0.0026	1.000	0.63	
zcat7	0.0044	0.0044	1.000	0.52	
zcat8	0.0057	0.0056	1.000	0.51	
zcat9	0.0057	0.0057	1.000	0.50	
zdt1	0.0032	0.0031	1.000	0.53	
zdt2	0.0030	0.0030	1.000	0.53	
zdt3	0.0015	0.0015	1.000	0.45	
zdt4	0.0008	0.0008	1.000	0.59	
zdt6	0.0031	0.0024	0.0001	1.00	✓

TABLE VI
NORMALIZED HYPERVOLUME EQUIVALENCE SUMMARY (PAIRED
BOOTSTRAP 90% CI; EQUIVALENCE MARGIN $\pm 1\%$ RELATIVE)

Framework	Eq. problems	Non-inf. problems	Median ΔHV (%)
pymoo	37/41	38/41	-0.01
jMetalPy	35/41	35/41	-0.02
DEAP	33/41	37/41	-0.01
Platypus	33/41	38/41	+0.05

TABLE VII
ROBUSTNESS SUMMARY FOR NORMALIZED HYPERVOLUME ACROSS
SEEDS (MEDIAN ACROSS PROBLEMS). “NEAR-BEST” IS DEFINED PER
PROBLEM AS $HV \geq 0.99 \cdot \max \text{MEDIAN}(HV)$ AMONG FRAMEWORKS

Framework	Median IQR(HV)	Median % seeds near-best
VAMOS	0.001	100.0
pymoo	0.001	100.0
jMetalPy	0.002	100.0
DEAP	0.001	100.0
Platypus	0.001	100.0

heatmap. The left panel compares VAMOS backends (Numba, MooCore, NumPy); the right panel compares VAMOS against pymoo and jMetalPy. Color intensity indicates runtime on a logarithmic scale, with annotated values in each cell and bold values marking the fastest per problem.

APPENDIX G STATISTICAL ANALYSIS FOR SMS-EMOA AND MOEA/D

Tables in this appendix report Wilcoxon signed-rank tests, bootstrap equivalence analyses, and robustness summaries for SMS-EMOA and MOEA/D, following the same methodology as the Statistical Analysis section of the main paper.

A. SMS-EMOA

Tables VIII–XI present per-problem Wilcoxon tests, equivalence summaries, and robustness summaries for SMS-EMOA. Fig. 5 visualizes per-problem bootstrap confidence intervals.

B. MOEA/D

Tables XII–XV and Fig. 6 present the corresponding analyses for MOEA/D.

APPENDIX H ALGORITHM CONFIGURATION SPACES

The full parameter space for NSGA-II with real-valued encoding is presented in the main paper.

APPENDIX I CODE EXAMPLES

Listing 1 shows how to autotune the NSGA-II tuning space with the Optuna backend. The `build_nsgaii_config_space` function defines the search space, which is converted to a `ParamSpace` before calling `ModelBasedTuner`.

NSGA-II — Equivalence CI (90%)

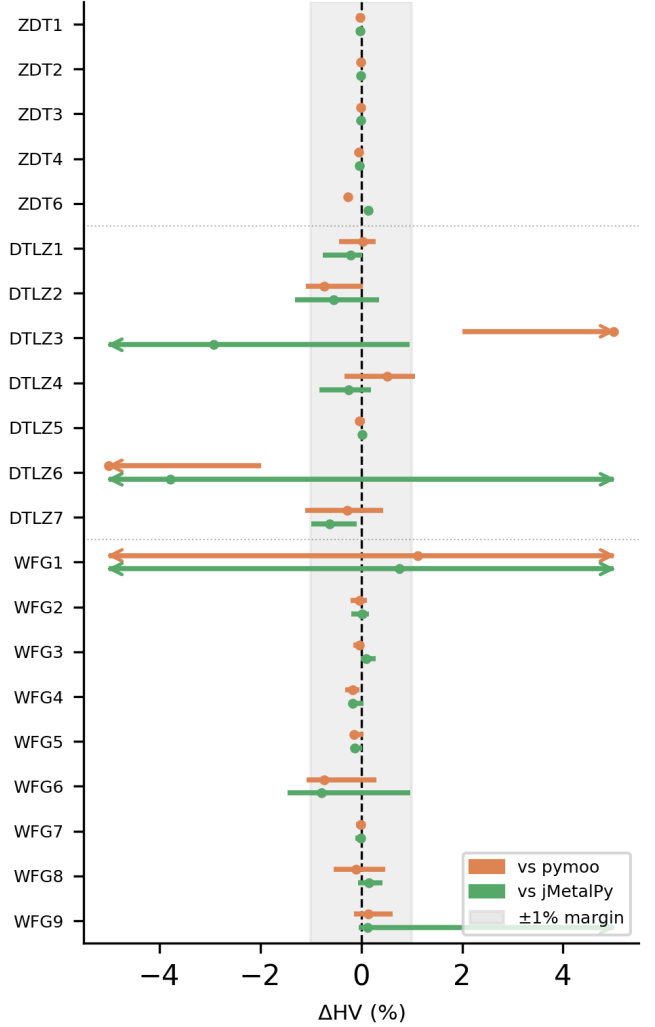


Fig. 3. NSGA-II. Per-problem paired bootstrap 90% confidence intervals for the relative normalized hypervolume difference $\Delta = (HV_{VAMOS} - HV_{fw}) / HV_{fw}$, reported in percent. The shaded band marks the $\pm 1\%$ equivalence margin; intervals falling entirely within this band indicate statistical equivalence

APPENDIX J PARAMETER SPACES FOR PYMOO AND JMETALPY

Tables XVI and XVII detail the configurable parameter spaces for NSGA-II in pymoo and jMetalPy, respectively.

REFERENCES

- [1] A. Vargha and H. D. Delaney, “A critique and improvement of the CL common language effect size statistics of McGraw and Wong,” *Journal of Educational and Behavioral Statistics*, vol. 25, no. 2, pp. 101–132, 2000.

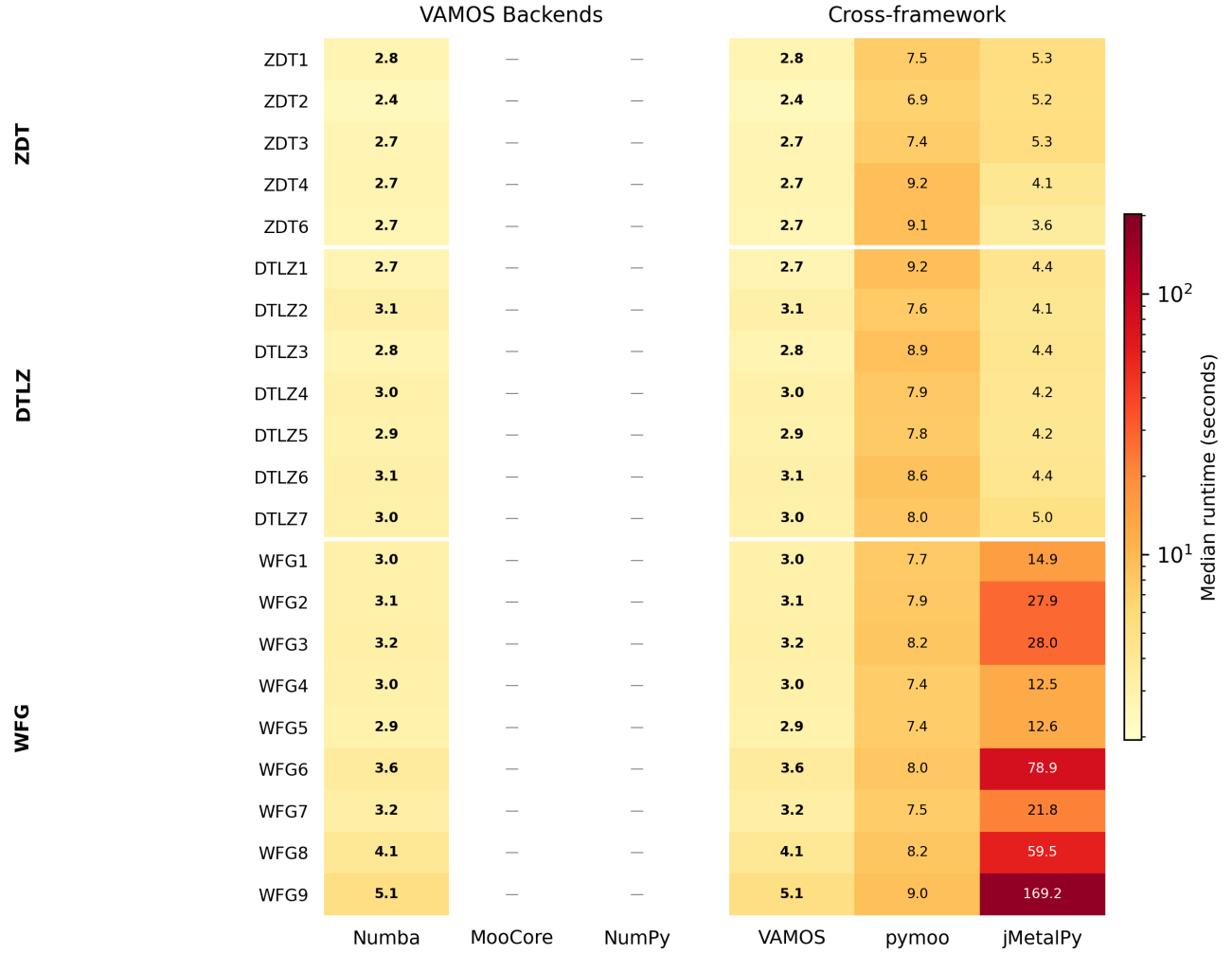


Fig. 4. Per-problem median runtime (seconds) for VAMOS backends (left) and cross-framework comparison (right). Color scale is logarithmic; bold values indicate the fastest in each row. Problems are grouped by family (ZDT, DTLZ, WFG).

Listing 1. Autotuning NSGA-II with Optuna on ZDT1, DTLZ2, and WFG4.

```

1 import numpy as np
2 from vamos import optimize, make_problem_selection
3 from vamos.engine.tuning import (
4     ModelBasedTuner, TuningTask, Instance, EvalContext,
5     build_nsgaii_config_space, config_from_assignment,
6 )
7 from vamos.foundation.quality_indicators import HVIndicator
8
9 # Evaluation function: returns hypervolume
10 def eval_fn(config, ctx: EvalContext):
11     cfg = config_from_assignment("nsgaii", config)
12     selection = make_problem_selection(
13         ctx.instance.name,
14         n_var=ctx.instance.n_var,
15         n_obj=ctx.instance.kwargs.get("n_obj"),
16     )
17     res = optimize(
18         selection.instantiate(),
19         algorithm="nsgaii",
20         algorithm_config=cfg,
21         max_evaluations=ctx.budget,
22         seed=ctx.seed,
23     )
24     hv = HVIndicator(reference_point=np.asarray(
25         ctx.instance.kwargs["ref_point"]
26     ))
27     return float(hv.compute(res.F).value)
28
29 # Define tuning task with one problem per family
30 task = TuningTask(
31     name="tune_nsgaii",
32     param_space=build_nsgaii_config_space().to_param_space(),
33     instances=[
34         Instance(name="zdt1", n_var=30,
35                 kwargs={"ref_point": [1.1, 1.1]}),
36         Instance(name="dtlz2", n_var=12,
37                 kwargs={"n_obj": 3,
38                         "ref_point": [1.1, 1.1, 1.1]}),
39         Instance(name="wfg4", n_var=24,
40                 kwargs={"n_obj": 2,
41                         "ref_point": [3.0, 5.0]}),
42     ],
43     seeds=[1, 2, 3], aggregator=np.mean,
44     budget_per_run=10000, maximize=True,
45 )
46
47 # Run Optuna-based tuning (TPE + multi-fidelity)
48 tuner = ModelBasedTuner(
49     task=task, max_trials=50, backend="optuna",
50     seed=42, n_jobs=4,
51     budget_levels=[2000, 5000, 10000],
52 )
53 best_config, history = tuner.run(eval_fn)

```

TABLE VIII
SMS-EMOA. NORMALIZED HYPERVOLUME COMPARISON WITH
WILCOXON SIGNED-RANK TEST RESULTS (HOLM-CORRECTED)

Problem	VAMOS	pymoo	p-value	$\hat{A}_{12}$	Sig.
zdt1	0.993	0.993	1.000	0.44	
zdt2	0.987	0.987	1.000	0.52	
zdt3	0.997	0.997	1.000	0.46	
zdt4	1.004	1.004	1.000	0.35	
zdt6	0.989	0.989	1.000	0.55	
dtlz1	0.959	0.959	1.000	0.51	
dtlz2	0.933	0.933	1.000	0.52	
dtlz3	0.902	0.901	1.000	0.50	
dtlz4	0.491	0.933	1.000	0.41	
dtlz5	0.978	0.978	0.003	0.20	✓
dtlz6	0.615	0.563	1.000	0.60	
dtlz7	0.949	0.950	0.092	0.17	
wfg1	0.210	0.258	1.000	0.40	
wfg2	1.009	1.010	1.000	0.54	
wfg3	0.984	0.983	1.000	0.52	
wfg4	0.980	0.980	1.000	0.51	
wfg5	0.836	0.836	1.000	0.47	
wfg6	0.854	0.858	1.000	0.44	
wfg7	0.982	0.982	1.000	0.35	
wfg8	0.773	0.775	1.000	0.34	
wfg9	0.961	0.961	1.000	0.46	
zcat1	0.991	0.991	1.000	0.51	
zcat2	0.996	0.996	0.008	0.75	✓
zcat3	0.991	0.991	1.000	0.41	
zcat4	0.991	0.991	1.000	0.41	
zcat5	0.997	0.997	1.000	0.49	
zcat6	0.998	0.998	1.000	0.36	
zcat7	0.995	0.995	1.000	0.49	
zcat8	0.985	0.985	1.000	0.36	
zcat9	0.985	0.985	1.000	0.36	
zcat10	0.977	0.977	1.000	0.54	
zcat11	0.995	0.995	1.000	0.60	
zcat12	0.994	0.994	1.000	0.36	
zcat13	1.020	1.020	1.000	0.62	
zcat14	0.988	0.988	1.000	0.47	
zcat15	0.994	0.994	1.000	0.36	
zcat16	0.994	0.994	1.000	0.47	
zcat17	0.997	0.997	1.000	0.50	
zcat18	0.995	0.995	1.000	0.49	
zcat19	0.993	0.993	1.000	0.47	
zcat20	0.995	0.995	1.000	0.48	

TABLE IX
SMS-EMOA, IGD+ COMPARISON WITH WILCOXON SIGNED-RANK TEST
RESULTS (HOLM-CORRECTED). LOWER IS BETTER

Problem	VAMOS	pymoo	p-value	$\hat{A}_{12}$	Sig.
zdt1	0.0023	0.0023	1.000	0.54	
zdt2	0.0022	0.0022	1.000	0.39	
zdt3	0.0010	0.0010	1.000	0.54	
zdt4	0.0003	0.0003	1.000	0.62	
zdt6	0.0019	0.0019	1.000	0.46	
dtlz1	0.0134	0.0134	1.000	0.48	
dtlz2	0.0184	0.0184	1.000	0.53	
dtlz3	0.0254	0.0257	1.000	0.50	
dtlz4	0.1994	0.0185	1.000	0.59	
dtlz5	0.0017	0.0017	1.000	0.51	
dtlz6	0.0730	0.0863	1.000	0.40	
dtlz7	0.0219	0.0217	0.760	0.72	
wfg1	0.9042	0.8573	1.000	0.59	
wfg2	0.2140	0.2138	1.000	0.46	
wfg3	0.0112	0.0115	1.000	0.48	
wfg4	0.0063	0.0062	1.000	0.54	
wfg5	0.0681	0.0681	1.000	0.58	
wfg6	0.0578	0.0552	1.000	0.55	
wfg7	0.0054	0.0053	1.000	0.59	
wfg8	0.1405	0.1388	1.000	0.62	
wfg9	0.0196	0.0218	1.000	0.45	
zcat1	0.0041	0.0041	1.000	0.48	
zcat2	0.0032	0.0032	1.000	0.47	
zcat3	0.0041	0.0041	1.000	0.50	
zcat4	0.0041	0.0041	1.000	0.57	
zcat5	0.0022	0.0022	1.000	0.53	
zcat6	0.0015	0.0015	1.000	0.58	
zcat7	0.0024	0.0024	1.000	0.50	
zcat8	0.0031	0.0031	1.000	0.40	
zcat9	0.0031	0.0031	1.000	0.41	
zcat10	0.0017	0.0017	1.000	0.48	
zcat11	0.0027	0.0027	1.000	0.44	
zcat12	0.0028	0.0028	1.000	0.60	
zcat13	0.0020	0.0020	1.000	0.51	
zcat14	0.0037	0.0037	1.000	0.47	
zcat15	0.0028	0.0028	1.000	0.63	
zcat16	0.0024	0.0024	1.000	0.46	
zcat17	0.0022	0.0022	1.000	0.51	
zcat18	0.0024	0.0024	1.000	0.49	
zcat19	0.0033	0.0033	1.000	0.42	
zcat20	0.0024	0.0024	1.000	0.55	

TABLE X
NORMALIZED HYPERVOLUME EQUIVALENCE SUMMARY (PAIRED
BOOTSTRAP 90% CI; EQUIVALENCE MARGIN $\pm 1\%$ RELATIVE)

Framework	Eq. problems	Non-inf. problems	Median Δ HV (%)
pymoo	39/41	39/41	-0.00
jMetalPy	36/41	38/41	+0.00

TABLE XI
ROBUSTNESS SUMMARY FOR NORMALIZED HYPERVOLUME ACROSS
SEEDS (MEDIAN ACROSS PROBLEMS). “NEAR-BEST” IS DEFINED PER
PROBLEM AS $HV \geq 0.99 \cdot \max \text{MEDIAN}(HV)$ AMONG FRAMEWORKS

Framework	Median IQR(HV)	Median % seeds near-best
VAMOS	0.000	100.0
pymoo	0.000	100.0
jMetalPy	0.000	100.0

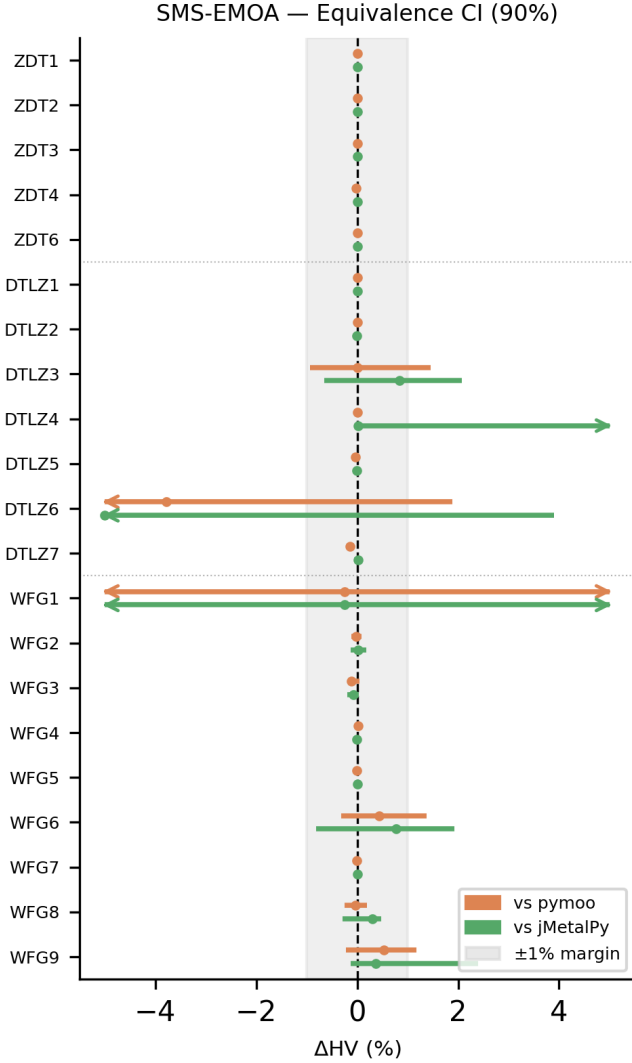


Fig. 5. SMS-EMOA. Per-problem paired bootstrap 90% confidence intervals for the relative normalized hypervolume difference $\Delta = (HV_{VAMOS} - HV_{fw})/HV_{fw}$, reported in percent. The shaded band marks the $\pm 1\%$ equivalence margin; intervals falling entirely within this band indicate statistical equivalence

TABLE XII
MOEA/D. NORMALIZED HYPERVOLUME COMPARISON WITH WILCOXON
SIGNED-RANK TEST RESULTS (HOLM-CORRECTED)

Problem	VAMOS	pymoo	p-value	$\hat{A}_{12}$	Sig.
zdt1	0.993	0.992	0.0001	0.98	✓
zdt2	0.986	0.986	0.010	0.74	✓
zdt3	0.993	0.993	0.0001	1.00	✓
zdt4	1.001	1.001	1.000	0.60	
zdt6	0.987	0.985	0.0001	0.98	✓
dtlz1	0.901	0.862	0.0001	1.00	✓
dtlz2	0.832	0.795	0.0001	1.00	✓
dtlz3	0.809	0.790	0.308	0.74	
dtlz4	0.833	0.637	0.034	0.86	✓
dtlz5	0.952	0.952	1.000	0.50	
dtlz6	0.602	0.601	1.000	0.47	
dtlz7	0.666	0.650	0.0001	0.85	✓
wfg1	0.241	0.373	0.0001	0.11	✓
wfg2	1.005	1.006	1.000	0.49	
wfg3	0.977	0.978	0.121	0.27	
wfg4	0.967	0.965	0.033	0.76	✓
wfg5	0.828	0.828	0.059	0.81	
wfg6	0.837	0.838	1.000	0.49	
wfg7	0.971	0.972	0.452	0.37	
wfg8	0.759	0.762	1.000	0.43	
wfg9	0.928	0.929	1.000	0.55	
zcat1	0.983	0.983	1.000	0.66	
zcat2	0.988	0.987	0.0001	0.98	✓
zcat3	0.983	0.983	1.000	0.63	
zcat4	0.983	0.983	1.000	0.57	
zcat5	0.991	0.991	0.0001	0.98	✓
zcat6	0.995	0.995	1.000	0.44	
zcat7	0.983	0.983	1.000	0.64	
zcat8	0.968	0.968	0.682	0.69	
zcat9	0.968	0.968	1.000	0.58	
zcat10	0.934	0.934	1.000	0.66	
zcat11	0.986	0.986	1.000	0.48	
zcat12	0.986	0.986	1.000	0.39	
zcat13	1.010	1.009	0.0001	1.00	✓
zcat14	0.976	0.976	0.190	0.75	
zcat15	0.986	0.986	1.000	0.45	
zcat16	0.982	0.982	1.000	0.50	
zcat17	0.991	0.991	0.0001	0.97	✓
zcat18	0.983	0.983	1.000	0.66	
zcat19	0.985	0.985	1.000	0.43	
zcat20	0.983	0.983	1.000	0.60	

TABLE XIII
MOEA/D, IGD+ COMPARISON WITH WILCOXON SIGNED-RANK TEST
RESULTS (HOLM-CORRECTED). LOWER IS BETTER

Problem	VAMOS	pymoo	p-value	$\hat{A}_{12}$	Sig.
zdt1	0.0025	0.0025	0.000	0.18	✓
zdt2	0.0025	0.0025	0.0001	0.12	✓
zdt3	0.0026	0.0027	0.0001	0.00	✓
zdt4	0.0008	0.0009	0.467	0.36	
zdt6	0.0023	0.0026	0.0001	0.01	✓
dtlz1	0.0218	0.0235	0.0001	0.11	✓
dtlz2	0.0334	0.0344	0.0001	0.01	✓
dtlz3	0.0395	0.0396	1.000	0.46	
dtlz4	0.0345	0.1186	0.020	0.13	✓
dtlz5	0.0036	0.0036	1.000	0.43	
dtlz6	0.0698	0.0748	1.000	0.48	
dtlz7	0.0746	0.0790	0.0001	0.07	✓
wfg1	0.9082	0.7454	0.0001	0.87	✓
wfg2	0.2169	0.2163	1.000	0.49	
wfg3	0.0171	0.0152	0.026	0.76	✓
wfg4	0.0102	0.0104	1.000	0.44	
wfg5	0.0694	0.0690	0.002	0.79	✓
wfg6	0.0746	0.0682	1.000	0.49	
wfg7	0.0094	0.0082	0.001	0.80	✓
wfg8	0.1382	0.1336	0.147	0.69	
wfg9	0.0385	0.0399	0.311	0.41	
zcat1	0.0070	0.0071	0.0001	0.11	✓
zcat2	0.0064	0.0068	0.0001	0.00	✓
zcat3	0.0072	0.0073	0.000	0.16	✓
zcat4	0.0072	0.0073	0.0001	0.14	✓
zcat5	0.0053	0.0055	0.0001	0.01	✓
zcat6	0.0027	0.0028	0.100	0.21	
zcat7	0.0061	0.0062	0.147	0.25	
zcat8	0.0066	0.0069	0.0001	0.06	✓
zcat9	0.0067	0.0068	0.000	0.14	✓
zcat10	0.0045	0.0046	0.0001	0.09	✓
zcat11	0.0065	0.0066	0.001	0.19	✓
zcat12	0.0065	0.0067	0.004	0.16	✓
zcat13	0.0060	0.0064	0.0001	0.00	✓
zcat14	0.0076	0.0078	0.0001	0.08	✓
zcat15	0.0066	0.0066	0.147	0.26	
zcat16	0.0068	0.0068	0.000	0.16	✓
zcat17	0.0053	0.0054	0.0001	0.01	✓
zcat18	0.0062	0.0063	0.311	0.23	
zcat19	0.0062	0.0063	0.001	0.14	✓
zcat20	0.0062	0.0062	0.147	0.26	

TABLE XIV
NORMALIZED HYPERVOLUME EQUIVALENCE SUMMARY (PAIRED
BOOTSTRAP 90% CI; EQUIVALENCE MARGIN $\pm 1\%$ RELATIVE)

Framework	Eq. problems	Non-inf. problems	Median ΔHV (%)
pymoo	32/41	39/41	+0.01
jMetalPy	36/41	38/41	+0.00
Platypus	20/20	20/20	+0.03

TABLE XV
ROBUSTNESS SUMMARY FOR NORMALIZED HYPERVOLUME ACROSS
SEEDS (MEDIAN ACROSS PROBLEMS). “NEAR-BEST” IS DEFINED PER
PROBLEM AS $HV \geq 0.99 \cdot \max \text{MEDIAN}(HV)$ AMONG FRAMEWORKS

Framework	Median IQR(HV)	Median % seeds near-best
VAMOS	0.000	100.0
pymoo	0.000	100.0
jMetalPy	0.000	100.0
Platypus	0.000	100.0

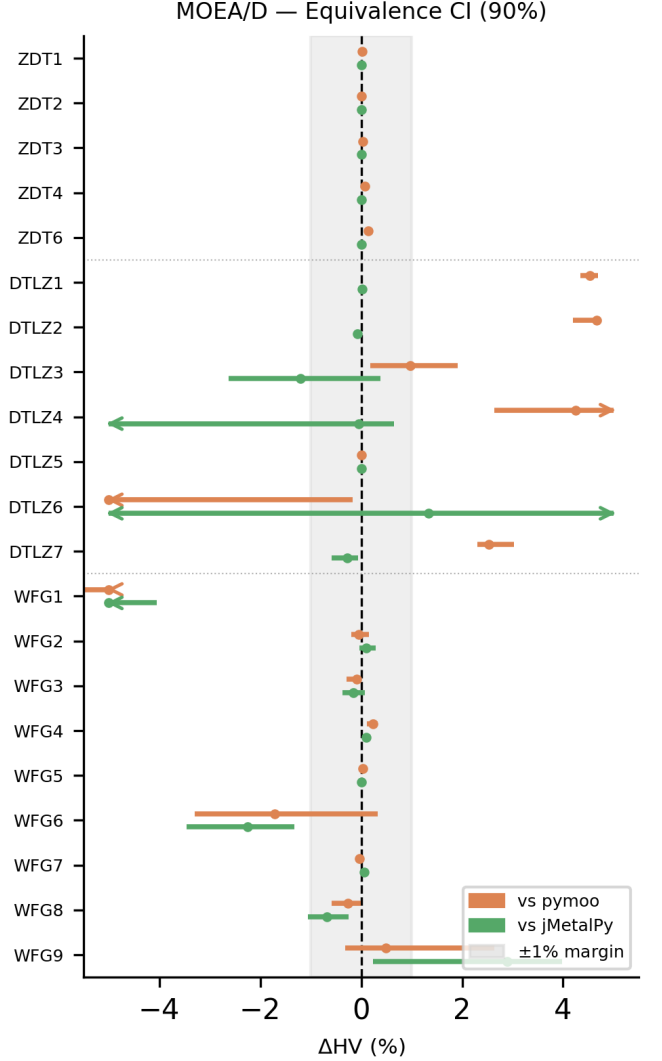


Fig. 6. MOEA/D. Per-problem paired bootstrap 90% confidence intervals for the relative normalized hypervolume difference $\Delta = (HV_{VAMOS} - HV_{fw})/HV_{fw}$, reported in percent. The shaded band marks the $\pm 1\%$ equivalence margin; intervals falling entirely within this band indicate statistical equivalence

TABLE XVI

PARAMETER SPACE FOR NSGA-II IN PYMOO (REAL-VALUED ENCODING). PARAMETERS IN ITALICS ARE CONDITIONAL, ACTIVE ONLY WHEN THEIR CORRESPONDING OPERATOR IS SELECTED. VALUES IN **BOLD** INDICATE DEFAULTS. THE TOTAL NUMBER OF CONFIGURABLE PARAMETERS IS 20 (11 ALWAYS ACTIVE, 9 CONDITIONAL).

Component	Value / Strategy	Conditional Parameters
General	Pop. Size: integer; default 100	–
	Offspring Count (<i>n_offsprings</i>): integer; default = Pop. Size	–
Initialization	{ Random , LHS}	–
Selection	{ Tournament , Random}	–
	<i>Tournament Size</i> (<i>pressure</i>): integer; default 2 Crowding Metric $\in \{\mathbf{cd}, \text{pcd}, \text{ce}, \text{mnn}, 2\text{nn}\}$	<i>Selection</i> = Tournament –
Crossover	<i>Global</i> : Probability $\in [0, 1]_{\mathbb{R}}$; default 0.9	<i>Dist. Index</i> $\eta \in [3, 30]_{\mathbb{R}}$; default 15 . <i>Per-var. prob.</i> $\in [0.1, 0.9]_{\mathbb{R}}$; default 0.5 $\eta \in [0.01, 0.3]_{\mathbb{R}}$, $\zeta \in [0.01, 0.3]_{\mathbb{R}}$ <i>CR</i> $\in [0, 1]_{\mathbb{R}}$; default 0.7. <i>F</i> : real or None
	SBX	
	PCX	
	DEX	
Mutation	<i>Global</i> : Prob. $\in [0, 1]_{\mathbb{R}}$; default 0.9 . <i>Per-var. prob.</i> ; default $\min(0.5, 1/n)$	<i>Dist. Index</i> $\eta \in [3, 30]_{\mathbb{R}}$; default 20 $\sigma \in [0.01, 0.25]_{\mathbb{R}}$; default 0.1
	Polynomial	
	Gaussian	
Repair	{ clip , bounce-back, to-bound}	–

TABLE XVII

PARAMETER SPACE FOR NSGA-II IN JMATALPY (REAL-VALUED ENCODING). PARAMETERS IN ITALICS ARE CONDITIONAL, ACTIVE ONLY WHEN THEIR CORRESPONDING OPERATOR IS SELECTED. VALUES IN **BOLD** INDICATE DEFAULTS. THE TOTAL NUMBER OF CONFIGURABLE PARAMETERS IS 29 (10 ALWAYS ACTIVE, 19 CONDITIONAL).

Component	Value / Strategy	Conditional Parameters
General	Pop. Size: integer	–
	Offspring Pop. Size: integer; = 1 for steady-state	–
	Use External Archive $\in \{\text{True}, \mathbf{False}\}$	–
	<i>Archive Type</i> $\in \{\text{NonDominated}, \text{CrowdingDist.}, \text{DistanceBased}\}$	<i>Archive</i> = True
	<i>Archive Max Size</i> : integer	<i>Archive</i> = True, bounded
Initialization	{ Random , Injector}	–
Selection	{ Binary Tournament , Tournament, Random}	–
	<i>Tournament Size</i> : integer ≥ 2	<i>Selection</i> = Tournament
Crossover	<i>Global</i> : Probability $\in [0, 1]_{\mathbb{R}}$	<i>Dist. Index</i> ; default 20 α ; default 0.5 α, β ; default 0.5 each – ζ ; default 0.5. η ; default 0.35 <i>CR</i> $\in [0, 1]_{\mathbb{R}}$, <i>F</i> $\in [0, 2]_{\mathbb{R}}$, <i>K</i> $\in [0, 1]_{\mathbb{R}}$
	SBX	
	BLX- α	
	BLX- $\alpha\beta$	
	Arithmetic	
	UNDX	
	DE	
Mutation	<i>Global</i> : Probability $\in [0, 1]_{\mathbb{R}}$; conventionally $1/n$	<i>Dist. Index</i> ; default 20 – <i>Perturbation</i> ; default 0.5 <i>Perturbation, Max Iterations</i> β ; default 1.5. <i>Step size</i> ; default 0.01 δ ; default 1.0
	Polynomial	
	Simple Random	
	Uniform	
	Non-Uniform	
	Lévy Flight Power Law	
Repair	{ Clamp , Random Uniform, Reflective, Bound Swap}	–